

Part: **3.1 – STRENGTH OF METALLIC PARTS**

Chapter: **3160**

Title: **SHEAR WEBS**

Revision: **0.1**

Date: **28/07/2021**

1 INTRODUCTION

This chapter describe the methodology proposed to perform the stress and strength analysis of metallic shear webs in beams, consisting of a flat plate with longitudinal axial stiffening members at its both sides, and subjected principally to shear and bending loads.

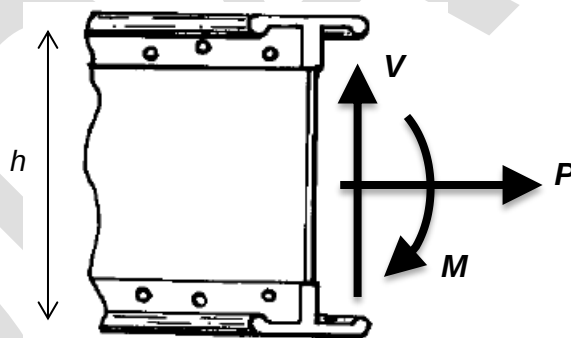


Figure 1. Sketch of typical shear web in a metallic beam

Webs of the kind treated here may have vertical stiffeners for preventing, or retarding, the buckling onset, but must be continuous, holes or other types of notches are not treated in this chapter.

2 BUCKLING RESISTANT WEBS

So referred those webs that are not permitted to buckling before ultimate loads.

2.1 STRESS DISTRIBUTION

2.1.1 Bending stress

The stress distribution in bending in metallic beams is discussed in chapter 3110 "BEAM SECTIONS". It is perfectly possible to use those general formulas for calculating the bending stress, in particular in buckling stable webs. Nevertheless, being the inertia of these types of beams quite concentrated in upper and lower caps/flanges, it is frequently possible to simplify the problem into a pair of forces case:

$$P_u = -P_l = M/h$$

Where:

P_u load force in upper flange/cap

P_l load force in lower flange/cap

M bending moment at section

h distance between the mass centres of the upper and lower caps/flanges

Then, with the areas of the flanges/caps the stresses at those points calculated, and assuming a linear distribution of strain across the web section, obtain a reasonable approximation of the stress distribution also in web.

2.1.2 Shear stress in web

The flexural shear stress problem is treated in chapter 3110 "BEAM SECTIONS".

In general, the webs of aircraft beams are relatively thin, compared with the area of the flanges/caps, and the shear stress over the web height is approximately uniform, being practical to simplify its calculation into the following expression:

$$f_s = \tau = V / (h \cdot t)$$

Where:

f_s, τ shear stress in web

V vertical shear force at section

h height of the web plate, frequently reduced to the distance between the mass centres of the upper and lower caps/flanges, or even between their riveting lines to the web

t web thickness

2.1.3 Beams with sloping flanges

A large majority of the beams in airplane wing and tail surfaces have sloping flanges because of the taper of the structure in both planform and thickness. This causes some coupling between bending reaction at flanges and web shear and must not be neglected. Assuming bending is reacted in axial loads by the sloping flanges the correction in vertical shear is as follows:

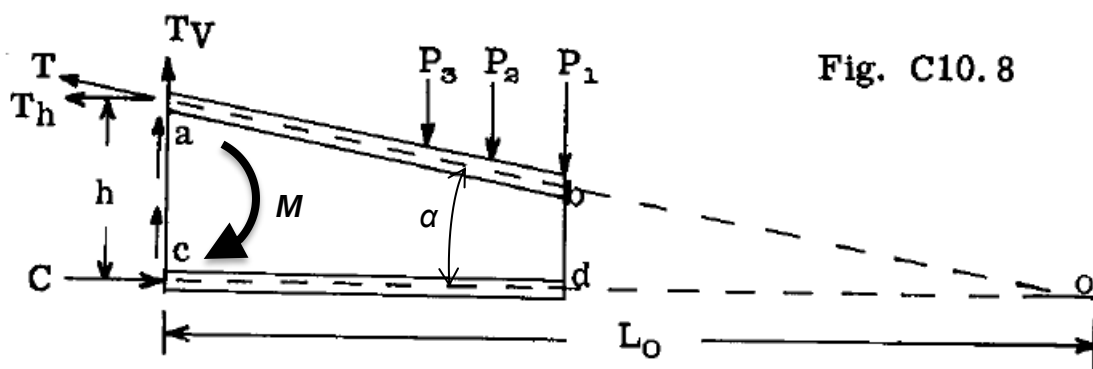


Figure 2. Bruhn's figure C10.8: webs with sloping flanges

Flanges loads and remaining bending moment at web:

$$C = T_h = M/h$$

$$T_v = (M/h)(h/L_0) = M/L_0 = (M/h) \cdot \sin(\alpha)$$

With the web effective in bending, the moment developed by the web should be subtracted from the total bending moment M .

$$M_{web} = M \cdot I_{web} / I$$

2.1.4 Strength

Axial stresses in flanges/caps and Von Mises stress in web shall be compared with material ultimate strength of the material to verify the strength of the (stable) web.

2.2 BUCKLING STABILITY

2.2.1 Local buckling of the web

Methods for calculating the buckling onset of flat rectangular plates are discussed in chapter 3510 "PLATE/SHELL BUCKLING". Fast/handbook methods and more sophisticated numerical methodologies can be used in different phases of the design, with different degrees of accuracy.

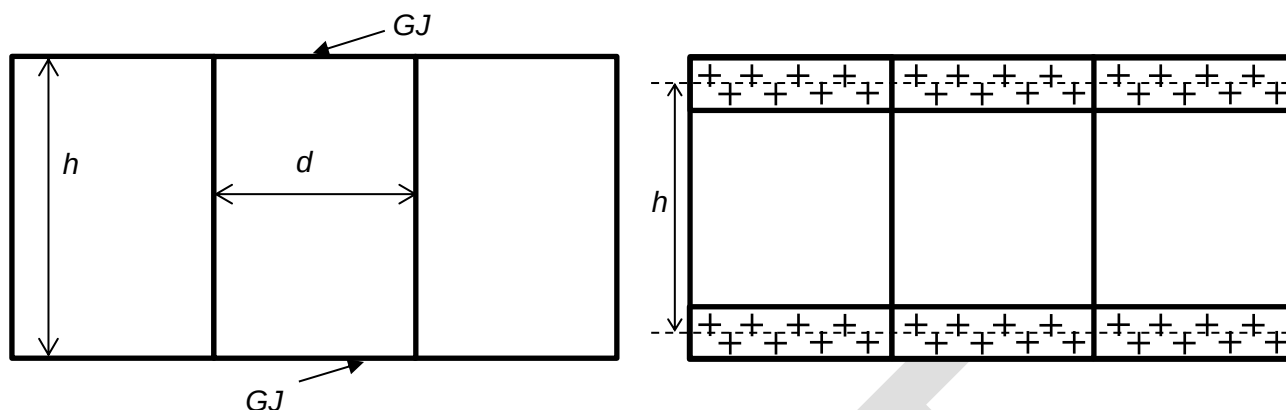


Figure 3. Dimensions of web subpanel in buckling analysis

In general simple support conditions on all the perimeter of the subpanels between vertical stiffeners shall be considered. The horizontal dimension of the panel for the buckling analysis shall be d , i.e. the distance between the midlines of the vertical stiffeners. The vertical dimension h for this analysis shall be the distance between the flanges, or the distance between the riveting midlines of the web attachment. Rotational stiffness of the flanges/caps can be used in analysis to account for more realistic support conditions.

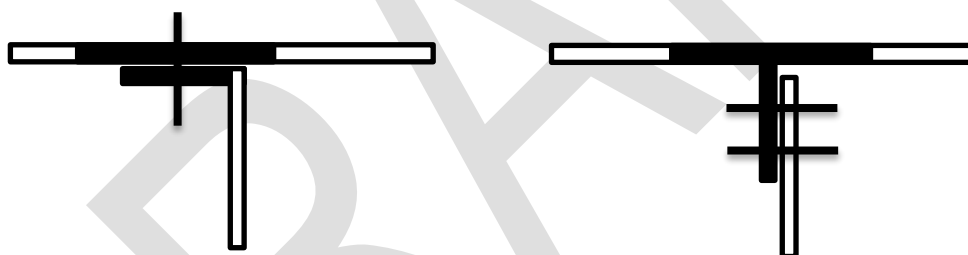


Figure 4. Areas of flanges proposed for GJ calculation as rotational constraint for web buckling analysis (shadowed segments with widths not greater than 10 times the thickness)

2.2.2 Effective vertical stiffeners

Instead of the expression proposed by H. Wagner, the following formula, expressed in terms of non-dimensional parameters and also, not load dependent, is preferred in order to evaluate if a stiffener section is capable to prevent global buckling in a shear web with vertical stiffeners:

$$I_{vst} / (d \cdot t^3) = 0.0217 / [d / (h \cdot K_s^{0.5})]^{8/3}$$

Bruhn Fig. C10.9

Where:

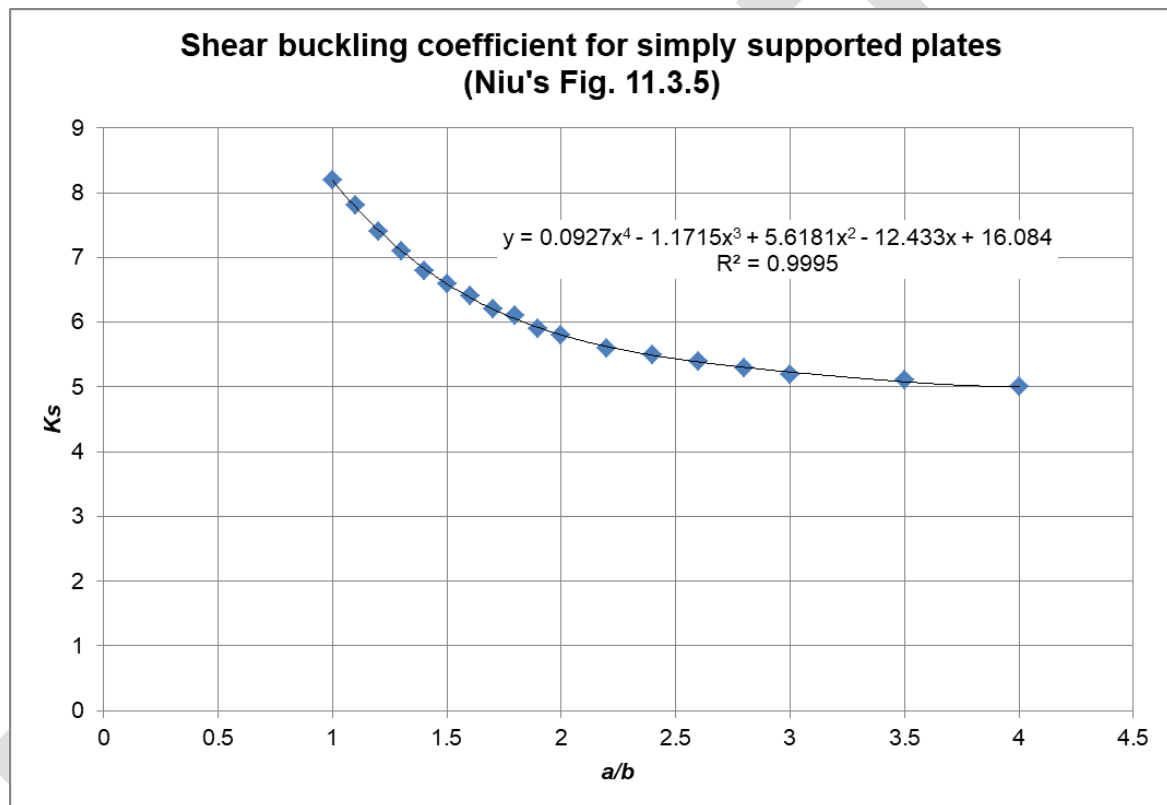
- I_{vst} moment of inertia of the stiffener (without eff. skin)
- d distance between stiffeners (d must be lower than h)
- h height of the web plate
- t web thickness

K_s Shear buckling coefficient (that in formulas for critical buckling shear stress $F_{s,cr} = K_s \cdot E \cdot [t/b]^2$, with b the minimum between h and d , thus d)

Expressed as an efficiency factor becomes:

$$F_{vst,eff} = 46 \cdot [I_v / (d \cdot t^3)] \cdot [d / (h \cdot K_s^{0.5})]^{8/3} \quad \dots \text{equal or greater than 1.0 to be effective}$$

It is strongly recommended, in particular in later phases of the design process, that the buckling stability analysis of the stiffened web is completed with more accurate numerical methodologies (e.g. FEM), in order to reduce conservatism of handbook predictions for the local buckling onsets, and to confirm buckling mode is local (and not global).



Note: Use ratio h/d instead of a/b in graph. If h/d is greater than 4, use $K_s = 5.0$.

Figure 5. Polynomial regression of the shear buckling coefficient in plates with hinged supports

2.2.3 Crippling strength of flanges/caps

Follow methodology discussed in chapter 3150 "LOCAL BUCKLING AND CRIPPLING OF BEAMS/STRINGERS" to verify the crippling strength of the flanges/caps under compression loads.

3 POST-BUCKLED WEBS

3.1 DIAGONAL TENSION

3.1.1 *Diagonal tension factor and angle*

The diagonal tension factor is obtained as follows (Kuhn, Ref. 3, eq. (27)):

$$k = \tanh [0.5 \cdot \log_{10}(\tau / \tau_{cr})] \quad \text{if } \tau > \tau_{cr}, \text{ otherwise } k = 0.0$$

alfa

The diagonal tension angle α can be generally assumed equal to 45° as a reasonably good approximation according to references consulted, even though, the process to get its value consistently with theory can be found in literature.

3.1.2 *Forced compression loads on stiffeners*

The forced compression on each longitudinal stiffener, induced by the diagonal tension at the adjacent subpanel subjected to shear N_{xy} , per Kuhn's eq. (51) (adding effective skin to the stringer section) will be expressed as:

$$\Delta P_x = -k \cdot N_{xy} \cdot b \cdot \cot(\alpha) = -k \cdot N_{xy} \cdot b / \tan(\alpha) \approx -k \cdot N_{xy} \cdot b$$

Similarly, on transversal stiffeners (Kuhn's eq. (52)):

$$\Delta P_y = -k \cdot N_{xy} \cdot a \cdot \tan(\alpha) \approx -k \cdot N_{xy} \cdot a$$

These compression forces induced by diagonal tension shall be added to primary loads in the stiffeners limiting the panel.

3.1.3 *Final loads in web*

The initial shear load in the panel q , after the calculation of the buckling onset, is treated separately to perform the (incomplete) diagonal tension analysis described in literature and summarized in previous section. From the diagonal tension angle α and factor k , the post-buckled loads, expressed in a reference system parallel to the diagonal tension direction can be calculated...

$$\begin{aligned}
 \begin{Bmatrix} N_x \\ N_y \\ N_{xy} \end{Bmatrix} &= \begin{Bmatrix} N_x \\ N_y \\ N_{xy} \end{Bmatrix}_{SR} + \begin{Bmatrix} N_x \\ N_y \\ N_{xy} \end{Bmatrix}_{DT} \\
 &= \begin{Bmatrix} (1-k)q \sin(2\alpha) \\ -(1-k)q \sin(2\alpha) \\ (1-k)q \cos(2\alpha) \end{Bmatrix}_{SR} + \begin{Bmatrix} 2kq / \sin(2\alpha) \\ 0 \\ 0 \end{Bmatrix}_{DT}
 \end{aligned}$$

Total Stress Resultant in Diagonal Tension Axis System = Resultant Due to Shear Resistant (SR) Action (Due to Pure Shear (1-k)q) + Resultant Due to Diagonal Tension (DT) Action (Due to Shear kq)

Figure 6. Final redistribution of load in skin in incomplete diagonal tension (Ref. 4)(*)

(*) Note the notation changes in figure:

- Original applied shear load N_{xy} is denoted as q ($q = \tau \cdot t$) in figure
- Additionally, N_k symbols in figure denote plate loads in the global reference axis system (axis parallel to panel sides) and N_k denote load in a local reference axis system with X parallel to the diagonal tension direction

4 APPENDICES

ABBREVIATIONS AND SYMBOLS

<i>Symbol</i>	<i>Units</i>	<i>Definition</i>
α	°	Diagonal tension angle
cr	-	As subscript, refers to critical (typically to buckling onset level)
d	mm	Distance between the centres of the vertical stiffeners
f_s, τ	MPa	Shear stress in web
F_{tu}	MPa	Ultimate tensile stress of the material
G	MPa	Shear modulus of the material
h	mm	Height/depth of the web
I	mm ⁴	Bending moment of inertia of a section
J	mm ⁴	Torsion constant/moment-of-inertia of a section
k	-	Diagonal tension factor
K_s	-	Shear buckling coefficient
M	Nmm	Bending moment in beam section
P	N	Axial load force
q	N/mm	In-plane shear load in panel
t	mm	Thickness of web
V	N	Vertical shear load force in beam section
l	-	As subscript, relative to the lower flange/cap
u	-	As subscript, relative to the upper flange/cap
s	-	As subscript, relative to shear
vst	-	As subscript, relative to the vertical stiffeners
w,web	-	As subscript, relative to the web

REFERENCES

	<i>Document Reference</i>	<i>Issue</i>	<i>Title</i>
Ref. 1	[M. C. Y. Niu]	2 nd Ed.	AIRFRAME STRESS ANALYSIS AND SIZING
Ref. 2	[E. F. Bruhn]	1973	ANALYSIS AND DESIGN OF FLIGHT VEHICLE STRUCTURES
Ref. 3	[P. Kuhn et al, NACA] TN-2661	1952	A SUMMARY OF DIAGONAL TENSION. PART I – METHODS OF ANALYSIS.
Ref. 4	[NASA] CR-97-20656	1997	Novel composites for wing and fuselage applications – speedy nonlinear analysis of postbuckled panels in shear (SNAPPS)

SOFTWARE

<i>Program</i>	<i>Version</i>	<i>Description</i>

Table 1. *Software programs referenced in this chapter*

RECORD OF REVISIONS

<i>Revision Date</i>	<i>Authors</i>	<i>Change Reason</i>	<i>Pages/Sub-chapters affected</i>
0.1 28/07/2021	G. Baños	First issue	All pages

Table 2. *Record of Revisions: authors, change reasons and sections/pages affected*

APPROVAL SIGNATURES TABLE

<i>Dpt. \ Site</i>	<i>Getafe</i>	<i>Manching</i>
<i>Stress Methods (TEPAP-TL4)</i>	G. Baños 28/07/2021	Florian Glock XX/XX/2021

Table 3. *Approval signatures table for last revision of this chapter*